

KALAIGAR ABDUL MUQEET

✉ am586288@gmail.com ☎ +91 9573437585 🌐 Abdul Muqeet 📁 Portfolio

SUMMARY

ADAS and Autonomous Systems Engineer with experience in embedded control, sensor integration, and ROS based autonomy for ground vehicles and production road vehicles. Worked on AEBS braking logic, vehicle longitudinal dynamics, and safety-critical decision-making. Strong background in robotics, decentralized navigation, multi-agent coordination, and AI/ML models using real-time sensor data.

EXPERIENCE

TIHAN IIT-Hyderabad - R&D, JRAF

10/2025 – Present
Hyderabad

- Contributed to autonomous navigation of a campus shuttle, implementing waypoint following, obstacle avoidance, path tracking, and motion control using ROS.
- Worked on Advanced Emergency Braking System (AEBS) functionality by integrating sensor-based distance estimation, vehicle speed, and braking decision logic.
- Assisted in vehicle platooning, enabling coordinated multi-vehicle movement, safe inter-vehicle spacing, and synchronized navigation.
- Performed system-level testing, data logging, and validation of autonomous driving behavior in controlled campus environments.
- Autonomous indoor UAV navigation in GPS-denied environments using RGB-D & visual-inertial cameras.
- Built indoor map using vision-based SLAM and enabled real-time localization and navigation for stable autonomous flight.
- Integrated perception and odometry for real-time obstacle avoidance during indoor UAV operations.

NEWRRO Tech LLP – Robotic Training

11/2024 – 12/2024
Bangalore, IN

- Engineered ESP32-based robotic systems and designed CAD models in Fusion 360.
- Integrated PLCs and control modules to build practical robotic solutions.

CRANES VARSITY – Post-Graduation Diploma in Data Science

08/2022 – 02/2023
Bangalore, IN

- Acquired python optimization and logic-building skills
- Applied programming to real world data structures and analysis
- SQL: Ability to work with relational databases and write SQL queries to interact with databases.

EDUCATION

ROBOTICS & AI (M. Tech), Nitte Meenakshi Institute of Technology
CGPA: 8.21

02/2024 – 08/2025
Bengaluru

ECE (B.E), HMS Institute of Technology
CGPA: 6.24

08/2018 – 02/2022
Tumkur

PROJECTS

Decentralized Navigation and Obstacle avoidance for Swarm Robots

- Developed a decentralized multi-robot navigation system with a master-slave framework, where follower robots autonomously tracked and followed a teleoperated master robot in real time.
- Implemented autonomous obstacle avoidance, localization, and path coordination using ROS navigation stack, TF transforms, and sensor fusion.

- Optimized swarm behavior to achieve reliable real-time following, cooperative movement, and scalability in multi-robot setups.

Optical Character Recognition Assisted Automated Medication Prescription Reader system

- Designed and implemented a full-stack web application to automatically extract structured medicine data from printed prescriptions using. Integrated Google Vision API for initial text detection and trained ML models to improve the accuracy of handwritten and printed text recognition.

Classification and Segmentation of Pneumonia medical condition using U-Net algorithm

- Enhanced a deep learning-based diagnostic system to classify chest X-ray images into normal and pneumonia-affected categories with high accuracy. Trained and fine-tuned CNN architectures for medical image classification tasks using labelled chest X-ray datasets. Implemented segmentation algorithms to isolate infected lung regions, where successfully achieved a 89% of Classification Accuracy.

Analyzing the data of Zomato Restaurants Data

- Conducted exploratory data analysis on a dataset of 12,000+ restaurants in Bengaluru to identify key factors influencing restaurant ratings, popularity, and customer preferences. Analyzed attributes such as cuisine type, location, average cost, delivery options, and online presence to determine their impact on customer engagement.

Air Pollution Monitoring System with SMS alert

- Designed and implemented a real-time air quality monitoring system using gas sensors to measure pollutants like CO₂, CO. Developed embedded software for continuous data acquisition and sensor calibration using microcontrollers.

SKILLS

Programming Languages

Python, C++ basics, SQL

Robotics

ROS, SLAM, Navigation Stack, LLMs

Machine Learning & AI

OpenCV, OCR, Deep Learning (CNNs), NLP, Pandas, NumPy, Scikit-Learn, TensorFlow/PyTorch

Soft Skills

Problem Solving, Research & Innovation, Adaptability, Communication

Tools & Platform

Git/GitHub, Linux, Docker, Fusion 360, Ansys, PLCs, UiPath RPA

PUBLICATIONS

Identification and Classification of Pneumonia medical condition using U-Net algorithm,

RCAAI 2025, Taylor & Francis

Accepted

Optical Character Recognition Assisted Automated Medication Prescription Reader system,

RCAAI 2025, Tylor & Francis

Accepeted

CERTIFICATIONS

- | | | |
|------------------------------|---|---|
| • UiPath Academy | • Udemy ROS2 for Beginners | • NPTEL: - Edge Computing, Big Data Computing |
| • Smart India Hackathon 2024 | • IEEE RAS Student Summit Participation | |